

Zepto Product & Inventory Analysis (SQL Project)

Project Overview

This project analyzes Zepto product dataset using PostgreSQL to explore pricing strategies, inventory distribution, discount patterns, and revenue opportunities. The analysis includes data cleaning, exploratory analysis, and business insight generation using SQL queries.

Data Cleaning

- Removed products with invalid pricing (MRP = 0)
- Converted price from paise to rupees
- Checked and handled NULL values

Key Analysis Performed

- Top products based on discount percentage
- Revenue and inventory value analysis by category
- Out-of-stock percentage and stock distribution
- Price per gram analysis for value comparison
- High-value and high-discount product identification
- Overpriced products detection using window functions

SQL Techniques Used

- Aggregations (SUM, AVG, COUNT)
- GROUP BY and HAVING
- Window Functions (AVG OVER PARTITION)
- Conditional aggregation (FILTER, CASE)
- Data cleaning operations (DELETE, UPDATE)
- Business logic implementation using SQL

Business Insights

- Identified categories with high inventory but low discount indicating potential overstock
- Found high-value products used for promotional strategies
- Highlighted out-of-stock trends across categories
- Detected pricing inconsistencies and overpriced products